

ee' of H is directed from e to e' in D . Then any stable matching in the graph $(X \cup Y, V(D'))$ for this set of preferences is a kernel in D' . $\square$

By Proposition 5.3.1, we now know the exact list-chromatic index of bipartite graphs: (5.3.1)

Corollary 5.4.5. *Every bipartite graph G satisfies $\text{ch}'(G) = \Delta(G)$.* $\square$

5.5 Perfect graphs

As discussed in Section 5.2, a high chromatic number may occur as a purely global phenomenon: even when a graph has large girth, and thus locally looks like a tree, its chromatic number may be arbitrarily high. Since such ‘global dependence’ is obviously difficult to deal with, one may become interested in graphs where this phenomenon does not occur, i.e. whose chromatic number is high only when there is a local reason for it.

Before we make this precise, let us define two new invariants for a graph G . The greatest integer r such that $K^r \subseteq G$ is the *clique number* $\omega(G)$ of G . The greatest size of a set of independent vertices in G is the *independence number* $\alpha(G)$ of G . Clearly, $\alpha(G) = \omega(\overline{G})$ and $\omega(G) = \alpha(\overline{G})$. $\omega(G)$
 $\alpha(G)$

A graph is called *perfect* if every induced subgraph $H \subseteq G$ has chromatic number $\chi(H) = \omega(H)$, i.e. if the trivial lower bound of $\omega(H)$ colours always suffices to colour the vertices of H . Thus, while proving an assertion of the form $\chi(G) > k$ may in general be difficult, even in principle, for a given graph G , it can always be done for a perfect graph simply by exhibiting some K^{k+1} subgraph as a ‘certificate’ for non-colourability with k colours. *perfect*

At first glance, the structure of the class of perfect graphs appears somewhat contrived: although it is closed under induced subgraphs (if only by explicit definition), it is not closed under taking general subgraphs or supergraphs, let alone minors (examples?). However, perfection is an important notion in graph theory: the fact that several fundamental classes of graphs are perfect (as if by fluke) may serve as a superficial indication of this.²

What graphs, then, are perfect? Bipartite graphs are, for instance. Less trivially, the complements of bipartite graphs are perfect, too – a fact equivalent to König’s duality theorem 2.1.1 (Exercise 41). The

² The class of perfect graphs has duality properties with deep connections to optimization and complexity theory, which are far from understood. Theorem 5.5.6 shows the tip of an iceberg here; for more, the reader is referred to Lovász’s survey cited in the notes.

so-called *comparability graphs* are perfect, and so are the *interval graphs* (see the exercises); both these turn up in numerous applications.

chordal

In order to study at least one such example in some detail, we prove here that the chordal graphs are perfect: a graph is *chordal* (or *triangulated*) if each of its cycles of length at least 4 has a chord, i.e. if it contains no induced cycles other than triangles.

pasting

To show that chordal graphs are perfect, we shall first characterize their structure. If G is a graph with induced subgraphs G_1 , G_2 and S , such that $G = G_1 \cup G_2$ and $S = G_1 \cap G_2$, we say that G arises from G_1 and G_2 by *pasting* these graphs together along S .

[12.3.6]

Proposition 5.5.1. *A graph is chordal if and only if it can be constructed recursively by pasting along complete subgraphs, starting from complete graphs.*

Proof. If G is obtained from two chordal graphs G_1, G_2 by pasting them together along a complete subgraph, then G is clearly again chordal: any induced cycle in G lies in either G_1 or G_2 , and is hence a triangle by assumption. Since complete graphs are chordal, this proves that all graphs constructible as stated are chordal.

Conversely, let G be a chordal graph. We show by induction on $|G|$ that G can be constructed as described. This is trivial if G is complete. We therefore assume that G is not complete, in particular that $|G| > 1$, and that all smaller chordal graphs are constructible as stated. Let $a, b \in G$ be two non-adjacent vertices, and let $X \subseteq V(G) \setminus \{a, b\}$ be a minimal a - b separator. Let C denote the component of $G - X$ containing a , and put $G_1 := G[V(C) \cup X]$ and $G_2 := G - C$. Then G arises from G_1 and G_2 by pasting these graphs together along $S := G[X]$.

a, b, X
 C
 G_1, G_2
 S

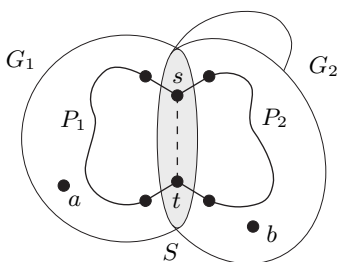


Fig. 5.5.1. If G_1 and G_2 are chordal, then so is G

Since G_1 and G_2 are both chordal (being induced subgraphs of G) and hence constructible by induction, it suffices to show that S is complete. Suppose, then, that $s, t \in S$ are non-adjacent. By the minimality of $X = V(S)$ as an a - b separator, both s and t have a neighbour in C . Hence, there is an X -path from s to t in G_1 ; we let P_1 be a shortest such

s, t

path. Analogously, G_2 contains a shortest X -path P_2 from s to t . But then $P_1 \cup P_2$ is a chordless cycle of length ≥ 4 (Fig. 5.5.1), contradicting our assumption that G is chordal. $\square$

Proposition 5.5.2. *Every chordal graph is perfect.*

Proof. Since complete graphs are perfect, it suffices by Proposition 5.5.1 to show that any graph G obtained from perfect graphs G_1, G_2 by pasting them together along a complete subgraph S is again perfect. So let $H \subseteq G$ be an induced subgraph; we show that $\chi(H) \leq \omega(H)$.

Let $H_i := H \cap G_i$ for $i = 1, 2$, and let $T := H \cap S$. Then T is again complete, and H arises from H_1 and H_2 by pasting along T . As an induced subgraph of G_i , each H_i can be coloured with $\omega(H_i)$ colours. Since T is complete and hence coloured injectively, two such colourings, one of H_1 and one of H_2 , may be combined into a colouring of H with $\max\{\omega(H_1), \omega(H_2)\} \leq \omega(H)$ colours – if necessary by permuting the colours in one of the H_i . $\square$

By definition, every induced subgraph of a perfect graph is again perfect. The property of perfection can therefore be characterized by forbidden induced subgraphs: there exists a set $\mathcal{H}$ of imperfect graphs such that any graph is perfect if and only if it has no induced subgraph isomorphic to an element of $\mathcal{H}$. (For example, we may choose as $\mathcal{H}$ the set of all imperfect graphs with vertices in $\mathbb{N}$.)

Naturally, one would like to keep $\mathcal{H}$ as small as possible. It is one of the deepest results in graph theory that $\mathcal{H}$ need only contain two types of graph: the odd cycles of length ≥ 5 and their complements. (Neither of these are perfect; cf. Theorem 5.5.4 below.) This fact, the famous *strong perfect graph conjecture* of Berge (1963), was proved only 40 years later:

Theorem 5.5.3. (Chudnovsky, Robertson, Seymour & Thomas 2006)
A graph G is perfect if and only if neither G nor $\overline{G}$ contains an odd cycle of length at least 5 as an induced subgraph.

*strong
perfect
graph
theorem*

In the context of perfect graphs, induced cycles of length at least 4 in G are usually referred to as *holes* in G , while holes in $\overline{G}$ are *antiholes* of G . In this jargon, the strong perfect graph theorem says that a graph is perfect if and only if it has neither odd holes nor odd antiholes.

*hole
antihole*

The proof of the strong perfect graph theorem is long and technical, and it would not be too illuminating to attempt to sketch it. Its overall design is not unlike our proof that chordal graphs are perfect: it decomposes the graph under consideration into simpler pieces, which are shown to be perfect first, in ways that preserve perfection when the pieces are put back together – much as the decomposition along complete subgraphs did in the case of chordal graphs.

To shed more light on the notion of perfection, we instead give two direct proofs of its most important consequence: the *perfect graph theorem*, formerly Berge's *weak perfect graph conjecture*:

perfect
graph
theorem

Theorem 5.5.4. (Lovász 1972)

A graph is perfect if and only if its complement is perfect.

We give two proofs for Theorem 5.5.4. The first is based on ideas from Lovász's original proof. The second is a magical linear algebra proof, due to Gasparian (1996), of a slightly stronger theorem of Lovász's.

The first proof of Theorem 5.5.4 is based on a crucial lemma. Let G be a graph and $x \in G$ a vertex, and let G' be obtained from G by adding a vertex x' and joining it to x and all the neighbours of x . We say that G' is obtained from G by *replicating* the vertex x to the pair $\{x, x'\}$ (Fig. 5.5.2).

replicating
a vertex

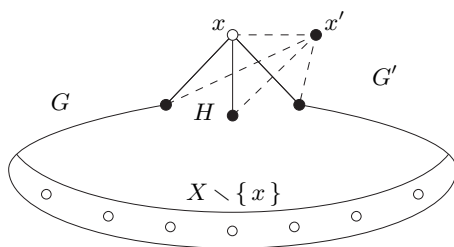


Fig. 5.5.2. Replicating the vertex x in the proof of Lemma 5.5.5

Lemma 5.5.5. (Replication Lemma)

Any graph obtained from a perfect graph by replicating a vertex is again perfect.

Proof. We use induction on the order of the perfect graph considered. Replicating the vertex of K^1 yields K^2 , which is perfect. For the induction step, let G be a non-trivial perfect graph, and let G' be obtained from G by replicating a vertex $x \in G$ to a pair $\{x, x'\}$. For our proof that G' is perfect it suffices to show $\chi(G') \leq \omega(G')$. Indeed, every *proper* induced subgraph H' of G' is either isomorphic to an induced subgraph H of G and hence perfect by assumption, or it is obtained from a proper induced subgraph of G by replicating x and hence perfect by the induction hypothesis. Either way, it can be coloured with $\omega(H')$ colours.

x, x'

ω

Let $\omega(G) = \omega$; then $\omega(G') \in \{\omega, \omega + 1\}$. If $\omega(G') = \omega + 1$, then

$$\chi(G') \leq \chi(G) + 1 = \omega + 1 = \omega(G')$$

and we are done. So let us assume that $\omega(G') = \omega$. Then x lies in no $K^\omega \subseteq G$: together with x' , this would yield a $K^{\omega+1}$ in G' . Let us colour G with ω colours. Since every $K^\omega \subseteq G$ meets the colour class X of x but

X

not x itself, the graph $H := G - (X \setminus \{x\})$ has clique number $\omega(H) < \omega$ (Fig. 5.5.2). Since G is perfect, we may thus colour H with $\omega - 1$ colours. Now X is independent, so the set $(X \setminus \{x\}) \cup \{x'\} = V(G' - H)$ is also independent. We can therefore extend our $(\omega - 1)$ -colouring of H to an ω -colouring of G' , showing that $\chi(G') \leq \omega = \omega(G')$ as desired. $\square$

Interestingly, while replicating a vertex preserves perfection, it does not preserve the property of $\chi = \omega$ if this holds only for the graph itself: we need that all induced subgraphs have this property too (Exercise 47).

Proof of Theorem 5.5.4. Given a perfect graph G , let us show that $\overline{G}$ is perfect too. We prove the following statement first:

Every induced subgraph H of G has a complete subgraph K that meets all its independent vertex sets of size $\alpha(H)$. (*)

For our proof of (*), let $\mathcal{A}$ be the set of all independent vertex sets of size $\alpha(H)$ in H . Let $\mathcal{A}' = \{A' \mid A \in \mathcal{A}\}$ be a set of disjoint copies of the sets in $\mathcal{A}$, one new set A' for each $A \in \mathcal{A}$, and let $U := \bigcup \mathcal{A}'$ be their disjoint union. For every vertex v of H write U_v for the set of its clones in U , one in each A' with $v \in A$.³ These sets U_v may be empty, but every vertex of H' lies in some U_v .

Define a graph H' on U by making U_v complete for every $v \in H$, and adding all $U_u - U_v$ edges whenever uv is an edge of H . The sets $A' \in \mathcal{A}'$ are independent in H' , because the $A \in \mathcal{A}$ are independent in H . And H' contains no larger independent sets than these A' . Indeed, any independent set in H' meets each U_v in at most one vertex; the vertices v whose U_v it meets are independent in H , so there are at most $\alpha(H) = |A'|$ such v ; and H' has no vertices outside the sets U_v . Since the A' are disjoint and all have the same size, the fact that H' contains no larger independent sets than these implies that it cannot be coloured with fewer than $|\mathcal{A}'|$ colours. Thus, $\chi(H') = |\mathcal{A}'|$.

Since H' arises from an induced subgraph of H by vertex replication, Lemma 5.5.5 implies that it is perfect. It thus has a complete subgraph K' of order $\chi(H')$. As $\chi(H') = |\mathcal{A}'|$ and K' has at most one vertex in each A' , it meets every $A' \in \mathcal{A}'$.

Let K be the subgraph of H induced by the vertices v for which K' meets U_v . This is a complete subgraph of H that meets every $A \in \mathcal{A}$, because K' meets every $A' \in \mathcal{A}'$. This completes the proof of (*).

Let us now show that $\overline{G}$ is perfect. Expressed in terms of G , we have to show that we can cover the vertices of any induced subgraph H of G by at most $\alpha(H)$ complete subgraphs K of H . These can be found inductively by $\alpha(H)$ applications of (*), since $\alpha(H - K) < \alpha(H)$ if $K \subseteq H$ satisfies (*). $\square$

³ This can be expressed formally by letting $A' := \{(v, A) \mid v \in A\}$ for every $A \in \mathcal{A}$, and $U_v := \{(v, A) \mid v \in A \in \mathcal{A}\} = \{(v, A) \in A' \mid A \in \mathcal{A}\}$ for every $v \in V(H)$.

The proof of the perfect graph theorem is best understood if we start from the last paragraph: once we have shown (*), the proof that $\overline{G}$ is perfect is easy.

The existence of a complete graph K as in (*) follows at once from the perfection of H if the sets in $\mathcal{A}$ provide an optimal colouring of H : if they cover its vertices and $|\mathcal{A}| \leq \chi(H)$. The first of these requirements, however, is immaterial. We can just replace H with its subgraph induced by those of its vertices that are covered by $\mathcal{A}$: that subgraph, too, will be perfect, and any K found in it will satisfy (*) also for the original H .

So assuming that $\mathcal{A}$ covers H , how can we prove that a colouring it induces is optimal? This would be immediate if the $A \in \mathcal{A}$ were disjoint: since they all have the same size, any colouring with fewer colours would need a larger colour class, which does not exist since the sets in $\mathcal{A}$ have size $\alpha(H)$. The trick, now, is simply to *make* the sets in $\mathcal{A}$ disjoint, as is done first thing in the proof of (*). The obvious way to copy the structure of H to the enlarged vertex set without creating any larger independent sets, then, is to endow it with the edges as in the definition of H' in the proof. This, in turn, motivates the notion of vertex replication.

All we need to complete the proof now is that the expanded graph H' is again perfect. This is Lemma 5.5.5.

Since the following characterization of perfection is symmetrical in G and $\overline{G}$, it clearly implies Theorem 5.5.4. As our proof of Theorem 5.5.6 will again be from first principles, we thus obtain a second and independent proof of the perfect graph theorem.

Theorem 5.5.6. (Lovász 1972)

A graph G is perfect if and only if

$$|H| \leq \alpha(H) \cdot \omega(H) \quad (*)$$

for all induced subgraphs $H \subseteq G$.

Proof. Let us write $V(G) = \{v_1, \dots, v_n\}$, and put $\alpha := \alpha(G)$ and $\omega := \omega(G)$. The necessity of (*) is immediate: if G is perfect, then every induced subgraph H of G can be partitioned into at most $\omega(H)$ colour classes each containing at most $\alpha(H)$ vertices, and (*) follows.

To prove sufficiency, we apply induction on $n = |G|$. Assume that every induced subgraph H of G satisfies (*), and suppose that G is not perfect. By the induction hypothesis, every proper induced subgraph of G is perfect. Hence, every non-empty independent set $U \subseteq V(G)$ satisfies

$$\chi(G - U) = \omega(G - U) = \omega. \quad (1)$$

Indeed, while the first equality is immediate from the perfection of $G - U$, the second is easy: ' $\leq$ ' is obvious, while $\chi(G - U) < \omega$ would imply $\chi(G) \leq \omega$, so G would be perfect contrary to our assumption.

Let us apply (1) to a singleton $U = \{u\}$ and consider an ω -colouring of $G - u$. Let K be the vertex set of any K^ω in G . Clearly,

$$\text{if } u \notin K \text{ then } K \text{ meets every colour class of } G - u; \quad (2)$$

$$\text{if } u \in K \text{ then } K \text{ meets all but exactly one colour class of } G - u. \quad (3)$$

Let $A_0 = \{u_1, \dots, u_\alpha\}$ be an independent set in G of size α . Let $A_1, \dots, A_\omega$ be the colour classes of an ω -colouring of $G - u_1$, let $A_{\omega+1}, \dots, A_{2\omega}$ be the colour classes of an ω -colouring of $G - u_2$, and so on; altogether, this gives us $\alpha\omega + 1$ independent sets $A_0, A_1, \dots, A_{\alpha\omega}$ in G . For each $i = 0, \dots, \alpha\omega$, there exists by (1) a $K^\omega \subseteq G - A_i$; we denote its vertex set by K_i .

Note that if K is the vertex set of any K^ω in G , then

$$K \cap A_i = \emptyset \text{ for exactly one } i \in \{0, \dots, \alpha\omega\}. \quad (4)$$

Indeed, if $K \cap A_0 = \emptyset$ then $K \cap A_i \neq \emptyset$ for all $i \neq 0$, by definition of A_i and (2). Similarly if $K \cap A_0 \neq \emptyset$, then $|K \cap A_0| = 1$, so $K \cap A_i = \emptyset$ for exactly one $i \neq 0$: apply (3) to the unique vertex $u \in K \cap A_0$, and (2) to all the other vertices $u \in A_0$.

Let $A = (a_{ij})$ be the real $(\alpha\omega + 1) \times n$ matrix whose rows are the incidence vectors of the sets A_i with $V(G)$: where $a_{ij} = 1$ if $v_j \in A_i$, and $a_{ij} = 0$ otherwise. Similarly, let B denote the real $n \times (\alpha\omega + 1)$ matrix whose columns are the incidence vectors of the sets K_j with $V(G)$. Now while $|A_i \cap K_i| = 0$ for all i by the choice of K_i , we have $A_i \cap K_j \neq \emptyset$ and hence $|A_i \cap K_j| = 1$ whenever $i \neq j$, by (4). Thus,

$$J := AB$$

is the $(\alpha\omega + 1) \times (\alpha\omega + 1)$ matrix with zero entries in the main diagonal and all other entries 1.

It is easy to see that J is non-singular, and hence has rank $\alpha\omega + 1$. But A , which has n columns, cannot have lower rank than J . Hence $n \geq \alpha\omega + 1$, which contradicts (*) for $H := G$. $\square$

5.6 χ -bounded graph properties

Our motivation for considering perfect graphs in the last section was that high chromaticity is easier to understand for these graphs than in general: if a perfect graph has chromatic number at least r , it has a local reason for this in the form of a K^r subgraph. It turned out that perfection has structural implications far beyond this. We may ask, therefore, whether graph properties defined by a slightly weaker condition might still imply some interesting structure for the graphs that have them.

χ -bounded
 χ -bounding

Consider any graph property $\mathcal{G}$ which, like perfection, is closed under taking induced subgraphs: a class that contains all induced subgraphs of its elements. We call $\mathcal{G}$ χ -bounded if there exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$, called a χ -bounding function, such that $\chi(G) \leq f(r)$ for all graphs $G \in \mathcal{G}$ of clique number $\omega(G) \leq r$, for all $r \in \mathbb{N}$. In particular, then, $\chi(G) \leq f(\omega(G))$; but note that for individual functions f this is a weaker condition. Perfection, for example, is χ -bounded with χ -bounding function $f(r) = r$.

Qualitatively, graphs in a χ -bounded class $\mathcal{G}$ have complete subgraphs as large as desired as soon as their chromatic number is big enough: that is the idea behind the notion of χ -boundedness. Quantitatively, if f is a χ -bounding function for $\mathcal{G}$, then any $G \in \mathcal{G}$ with $\chi(G) > f(r)$ contains a K^{r+1} , for every $r \in \mathbb{N}$.

Not all graph classes that are closed under taking induced subgraphs are χ -bounded: the triangle-free graphs, for example, are not, by Erdős's Theorem 5.2.5. So which classes are?

Every graph property $\mathcal{G}$ that is closed under induced subgraphs is characterized by the class $\mathcal{H}$ of graphs that are not in $\mathcal{G}$ and are minimal with this property under the induced-subgraph relation: an arbitrary graph G lies in $\mathcal{G}$ if and only if it has no induced subgraph in $\mathcal{H}$. One way to search for χ -bounded graph properties, therefore, is to examine those characterized in this way by some particularly simple classes $\mathcal{H}$.

The strong perfect graph theorem implies that $\mathcal{G}$ is χ -bounded when $\mathcal{H}$ consists of all odd holes and antiholes. But since χ -boundedness is weaker than perfection, we can make $\mathcal{H}$ smaller, thereby making $\mathcal{G}$ larger:

Theorem 5.6.1. (Scott & Seymour 2016)
The graphs with no odd hole are χ -bounded.

Theorem 5.6.1 is one of the earliest results of its kind proved in the wake of the strong perfect graph theorem. It has since been strengthened in various ways, see the notes. The notion of χ -boundedness itself is much older. It was introduced by Gyárfás in 1975, who later proved the following pretty result:

Proposition 5.6.2. *If P is any fixed path, then the graphs not containing P as an induced subgraph are χ -bounded.*